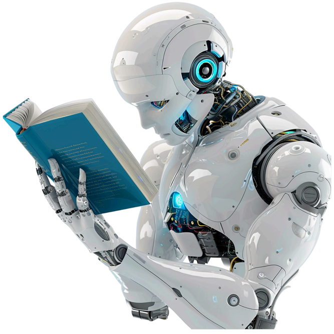

Midterm 2

General Instructions

In this second midterm, your goal is to explore, structure, model, and interpret a complex real-world dataset using both supervised and unsupervised learning techniques.

This assignment evaluates:

- Your ability to extract structure from raw data
- Your skill in model selection and tuning
- Your understanding of ensemble learning and probabilistic classification
- Your capacity to interpret models, not just report metrics

If you are retaking this test, you must use different datasets. Failing to abide by this rule will result in 0 points.

Academic Integrity & Use of AI Tools – Mandatory Notice

This project must represent **your own independent work**.

Strictly prohibited:

- Submitting **AI-generated code** (including code produced by ChatGPT, Copilot, or similar tools).
- Copying code, notebooks, or solutions from **GitHub, Kaggle notebooks, blogs, or any other online sources**.
- Any form of **plagiarism, collaboration, or unauthorized assistance**.

Sanctions for violations:

Any violation of academic integrity will result in **severe disciplinary measures**, including:

- **Immediate annulment of all points obtained in this course so far**,
- **Failing the course**, and
- **Formal disciplinary procedures**, which may lead to **suspension or expulsion from the university**, in accordance with university regulations.

Code similarity detection and manual inspection will be applied.

If you use external sources **for learning**, they must be **properly cited**, and all submitted code must be **written and fully understood by you**.

Dataset Rules

You must select **one Kaggle dataset** that:

- Is **not included in scikit-learn**
- Has **at least 2,000 samples**
- Contains **at least 10 meaningful features**
- Allows for **both clustering and classification analysis**

You must provide a **direct Kaggle link**.

The dataset should be:

- Rich enough to support **unsupervised exploration**
- Suitable for **classification tasks**
- Non-trivial (not toy datasets, not overly clean, not artificially simple)

You may:

- Clean and preprocess the dataset
- Engineer features
- Perform dimensionality reduction (optional)

Part I — Unsupervised Exploration & Structure Discovery (10 points)

Using **k-Means and Hierarchical Clustering**, you must explore the dataset and identify **latent structure**.

Required tasks:

- Perform **exploratory data analysis (EDA)**
- Apply **k-Means clustering**:
    - Determine a suitable number of clusters
    - Justify your choice
- Apply **Hierarchical clustering**:
    - Analyze dendrogram structure
    - Compare cluster consistency with k-Means

You must analyze:

- What do these clusters represent?
- Are they meaningful in real-world terms?
- Which features drive the separation?
- What does clustering reveal that raw inspection does not?

Part II — Supervised Learning & Model Competition (14 points)

From the same dataset, define a **classification problem**.

You must train and compare:

- **Support Vector Machine (SVM)**
- **Random Forest**
- **Boosting method (AdaBoost or Gradient Boosting)**
- **Naive Bayes classifier**

Required elements:

- K-fold cross-validation
- Hyperparameter tuning for **SVM and ensemble models**
- Performance evaluation using:
  - Precision
  - Recall
  - ROC
  - AUC

Key Analytical Focus

You must go beyond raw metrics and discuss:

- Why certain models perform better
- Bias—variance behavior
- Stability vs performance tradeoffs
- Interpretability vs predictive power
- Probabilistic reasoning using Naive Bayes

Part III — Model Interpretation & Trust (6 points)

This section focuses on **model understanding**, not performance.

You must analyze:

- Feature importance in ensemble models
- Decision boundaries of SVM
- Probabilistic outputs of Naive Bayes
- Model confidence and uncertainty

You must discuss:

- Which model you would trust most in a real deployment scenario
- Which model you would *not* trust — and why
  Where models fail and what that reveals about the dataset

Final Reflection Essay (Mandatory — 1 A4 Page, Font 12)

You must submit a **separate analytical reflection** of **exactly one A4 page (font size 12)** discussing:

- What clustering revealed about the data
- How unsupervised learning influenced your supervised modeling choices
- Strengths and weaknesses of ensemble methods vs SVM
- When probabilistic models are preferable
- What you learned about real-world machine learning workflows

Deliverables

1. Jupyter Notebook (.ipynb)

Must contain:

- Clean and structured code
- Clear explanations
- Visualizations
- Logical narrative flow

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2. Technical Report (PDF, 2–4 pages)

Must include:

- Dataset description
- Clustering analysis
- Model comparison
- Interpretation of results
- Final conclusions